BEYOND
EVALUATION
USE

An Empirical Study of
Involvement and Use/Influence in
Multi-Site Evaluations

Report 9:
NSF Second Follow-Up Survey Detailed Data Report

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Survey Description

The Beyond Evaluation Use Grant (EUG) team acquired names and email addresses for the principal investigators (PIs) and co-principal investigators (co-PIs) for each of the three NSF programs: ATE, MSP, and Noyce. Descriptions of the programs and their respective evaluations are provided in other grant-related documents. The team downloaded the lists from the NSF website. When names were missing email addresses, the EUG team searched the Internet for an email address. The total population for this study consisted of all 763 project PIs and co-PIs listed on the NSF website.

Online Survey

An online survey was conducted with project leaders in program-level evaluations of three large, multi-site science, technology, mathematics and engineering education interventions.

The PIs and co-PIs received an initial invitation email with a description of the survey, an explanation of its use, the Human Subjects approval number, and telephone numbers to call with questions. The email also contained a link to the online survey with a login and password. Two and five weeks following the first email, the team sent email reminders to non-respondents.

Prior to the initial invitation, researchers sent out pre-notice emails to approximately half of the PIs and co-PIs from each of the three programs. Recipients of the pre-notice email were selected through random sorting in Microsoft Excel. Due to uneven distribution of undeliverable emails, while half of the potential respondents in each program were selected to receive the pre-notice, not all of those selected actually received the email.

The EUG team staggered the distribution of the survey emails by program to minimize technical difficulties associated with mass emails (e.g., returned emails, server overload). Although distribution was staggered, all programs received an initial invitation email with a description of the survey, an explanation of its use, and the Human Subjects approval number and phone numbers to call with questions. The email also contained a link to the online survey with a login and password. Two and five weeks following the first email, the team sent reminders to non-respondents. The data were collected from April 8, 2009 through May 31, 2009.

Response Rate

Due to the fact that many of the email addresses were obtained via Internet search by the EUG team, only nine percent (68/763) of the emails were returned as undeliverable. After receiving the initial email invitation, a number of potential respondents replied with valid reasons they should not be included in the survey, such as those on projects that had not yet begun the evaluation process or those who retired prior to their project being evaluated. These persons were removed from the sample. The sampling frame, the number of ATE, MSP, and Noyce PIs and co-PIs with valid email addresses, was 669, with an overall response rate of 45% (300/669).

Respondents came from all but nine states in the U.S (Alaska, Delaware, Kansas, Nebraska, New Hampshire, North Dakota, Rhode Island, Utah, and Wyoming), as well as the District of Columbia. Fifty-five percent of the sample was male, 42% female, and three percent preferred not to respond.

Table 1. Surveys Completed

Program	**Respondents**	**Response Rate**		**Removed from Sample**
ATE	163	43%	163/381	52
MSP	41	45%	41/91	17
Noyce	96	49%	96/197	25
Total	300	45%	300/669	94

Pre-notice Email Analysis

The research team tested the use of a pre-notice email and its relation to response rates. For each of the three programs, response rates were higher for those respondents who had received a pre-notice email. For Noyce program respondents, a chi square analysis showed that there is a statistically significant relationship between receipt of a pre-notice email and response to the survey. When all programs were considered in total, this statistically significant relationship also existed.

Table 2. Surveys Completed Based on Receipt of Pre-Notice Email

	Pre-Notice Email			**No Pre-Notice Email**		
Program	**Respondents**	**Response Rate**		**Respondents**	**Response Rate**	
ATE	89	47%	89/189	75	39%	75/192
MSP	21	50%	21/42	20	40%	20/50
Noyce	55	57%	55/97	40	39%	40/102
Total	165	50%	165/328	135	39%	135/345

Table 3. Pre-Notice Email Chi Square Analysis

Program	Pearson χ^2	df	p-value	Cramer's φ
ATE	2.640	1	.104	.083
MSP	.924	1	.336	.100
Noyce	6.093	1	**.014**	.175
Total	8.498	1	**.004**	.112

EUG Survey Non-response Study Summary Data

To check for non-response error, the research team conducted a non-respondent follow-up survey following the online survey of the PIs and co-PIs in each of the three programs. Non-response error results when the people who responded to the survey are different from the individuals in the sample who did not respond, in a way that is relevant to the study. We developed a three-question non-respondent survey to address the topics of involvement and use, as well as to gather some information about what kept individuals from responding to the initial survey and reminder emails. Recipients of the non-respondent email were asked to select from the following responses: 1 – not at all involved; 2 – involved a little; 3 – involved some; and 4 – involved extensively. The non-respondent survey items were:

1. How involved were you in the *[Name of Program Evaluation]*?
2. How much impact did the *[Name of Program Evaluation]* have on you?
3. What was the main thing that kept you from responding to our initial survey request?

A random sample of non-respondents to the survey in each of the three programs (ATE, MSP, and Noyce) was drawn using Microsoft Excel. Each person was sent a personal, individually addressed email with a request for their input on the three items. Two reminder emails were sent to each of the respondents who did not respond. The table below shows the response rates of both the initial survey ("respondents") and the non-respondent follow-up survey ("non-respondents").

Table 4. Response Rates

	Respondents		Non-Respondents	
ATE	43%	163/381	32%	24/74
MSP	45%	41/91	47%	14/30
Noyce	49%	96/197	63%	22/35
Total	45%	300/669	43%	60/139

The results of the analysis comparing the respondents and non-respondents showed differences in the overall levels of involvement and use reported. In each case the mean of the survey respondents was higher than the mean of the non-respondents. The Involvement Mean for the respondents was calculated as the mean of thirteen questions on the original survey. The Use/Impact Mean was calculated in the same way, using the fourteen survey questions related to evaluation use and impact. The means for the non-respondents were simply the means of their answers to the two questions they responded to via email. Though these measurements are not identical, the two sets of means closely mirror one another in that the 4-point scale for each question or set of questions is the same. The differences

shown in the table below may indicate the possibility of upward bias in the estimates of overall involvement and use/impact for survey respondents or it may also be an artifact of the difference between use, influence and overall impact.

Table 5. Summary of Findings of Mean Comparisons

	Involvement Mean		Use/Impact Mean	
Program	**Respondents**	**Non-Respondents**	**Respondents**	**Non-Respondents**
ATE	2.60	1.56	2.66	1.56
MSP	2.24	1.71	2.54	1.71
Noyce	1.99	1.36	1.98	1.32

Another way to determine the similarity of survey respondents and non-respondents is to compare them on known variables. To compare the respondents with the non-respondents (non-respondents in this comparison include all those who received the original invitation for the survey, but did not complete the survey, not just those who were sent the non-respondent email) the research team conducted chi square analyses of two demographic items that were available from the original project lists downloaded from the NSF website – role and project start date. Analyses were conducted to see if it was statistically significantly more likely that a respondent or non-respondent fell into different categories, including PI vs. co-PI and the year in which the projects began. Findings show that, in all but 1 case, there was no statistically significant relationship between being a respondent and being a PI or being a respondent and having one's project begin in a given year. Though the Noyce project did show a significant relationship between respondents and PI status, it did not show a significant relationship between respondents and the start date of the projects. Overall, using these demographic data, it appears that the respondents and non-respondents are relatively similar. Results of the chi square analysis are included in the table below.

Table 6. ATE Respondent vs. Non-respondent and PI vs. Co-PI Percentages

	PI	Co-PI
Respondent	48%	40%
Non-respondent	52%	60%

Pearson χ^2 (1, n=379) = 2.290, p = .130, Cramer's φ = .078

Table 7. MSP Respondent vs. Non-respondent and PI vs. Co-PI Percentages

	PI	Co-PI
Respondent	40%	49%
Non-respondent	60%	51%

Pearson χ^2 (1, n=91) = .821, p = .365, Cramer's φ = .095

Table 8. Noyce Respondent vs. Non-respondent and PI vs. Co-PI Percentages

	PI	Co-PI
Respondent	67%	38%
Non-respondent	33%	62%

Pearson χ^2 (1, n=194) = 16.297, **p <.001**, Cramer's φ = .286

Table 9. ATE Respondent vs. Non-respondent and Project Start Year Percentages

	2003	2004	2005	2006	2007
Respondent	44%	43%	34%	50%	41%
Non-respondent	56%	57%	66%	50%	59%

Pearson χ^2 (4, n=379) = 5.347, p =.253, Cramer's φ = .119
*Note – unable to complete this chi square analysis for MSP due to small expected cell counts

Table 10. Noyce Respondent vs. Non-respondent and Project Start Year Percentages

	2003	2004	2005	2006	2007
Respondent	40%	42%	54%	51%	46%
Non-respondent	60%	58%	46%	49%	54%

Pearson χ^2 (4, n=194) = 1.963, p =.743, Cramer's φ = .101

Reasons for Not Responding

Of the 59 total responses to the non-respondent study across all three programs, 48% reported that they did not feel the survey applied to them, and nearly 12% did not remember having received the initial survey. In addition, approximately 25% cited a lack of time. Over 15% offered a range of specific reasons in the "other" category, the basic themes of which were: a) they felt the survey was too long b) they felt the survey was unanswerable or didn't understand the questions; c) they were retired or otherwise no longer involved in the project and d) a variety of individual explanations, such as family emergencies or sabbatical leaves. No one reported having had technical problems.

Report Format

For the purposes of this report, data from all cases are presented separately for each program (ATE, MSP, and Noyce). This format allows a reader to view results from all programs individually, which is most appropriate given the differences in both program and program evaluation design.

ATE Responses

Table 11. Surveys Completed – ATE

Respondents	Response Rate		Removed from Sample
163	44%	163/369	52

Table 12. Evaluation Experience – ATE

First Evaluation	Novice	Skilled	Expert	Total
43	47	59	10	159
27.0%	29.6%	37.1%	6.3%	100.0%

Involvement – ATE

Thirteen of the first 16 questions on the survey asked respondents about their perceptions of involvement during the planning, implementation and communication phases of the program evaluation. Each involvement question began with "I was involved in…." This section included a question at the end of each phase regarding the motivation for involvement. Results of the motivation questions (4, 11, & 16) are presented after the Involvement tables.

Table 13. ATE Involvement – Planning

Items 1-3	**No**	**Yes, A little**	**Yes, Some**	**Yes, Extensively**	**N**
Discussions that focused the evaluation	14.6%	8.9%	24.8%	51.6%	157
Identifying evaluation planning team members	29.9%	9.7%	18.2%	42.2%	154
Developing the evaluation plan	26.5%	10.3%	28.4%	34.8%	155

Table 14. ATE Involvement – Implementation

Items 5-10	**No**	**Yes, A little**	**Yes, Some**	**Yes, Extensively**	**N**
Developing data collection instruments	34.0%	17.3%	23.1%	25.6%	156
Developing data collection processes	29.0%	18.1%	25.2%	27.7%	155
Collecting data	33.8%	15.9%	18.5%	31.8%	157
Reviewing collected data for accuracy and/or completeness	37.6%	8.3%	21.0%	33.1%	157
Analyzing data	38.7%	11.0%	27.7%	22.6%	155
Interpreting collected data	29.9%	11.0%	29.9%	29.2%	154

Table 15. ATE Involvement – Communications

Items 12-15	**No**	**Yes, a little**	**Yes, some**	**Yes, extensively**	**N**
Writing evaluation reports	50.0%	15.6%	16.2%	18.2%	154
Reviewing evaluation reports for accuracy and/or completeness	23.5%	17.0%	24.8%	34.6%	153
Presenting evaluation findings (e.g., to staff, to stakeholders, to an external audience)	33.3%	7.8%	22.9%	35.9%	153
Developing future project plans based on evaluation results	16.6%	9.9%	23.8%	49.7%	151

Table 16. Involvement - ATE

Items 1-3, 5-10, 12-15	Mean	Median	N	SD
Discussions that focused the evaluation	3.13	4.00	157	1.087
Identifying evaluation planning team members	2.73	3.00	154	1.285
Developing the evaluation plan	2.72	3.00	155	1.199
Developing data collection instruments	2.40	2.00	156	1.201
Developing data collection processes	2.52	3.00	155	1.181
Collecting data	2.48	3.00	157	1.254
Reviewing collected data for accuracy and/or completeness	2.50	3.00	157	1.294
Analyzing data	2.34	3.00	155	1.208
Interpreting collected data	2.58	3.00	154	1.197
Writing evaluation reports	2.03	1.50	154	1.182
Reviewing evaluation reports for accuracy and/or completeness	2.71	3.00	153	1.175
Presenting evaluation findings (e.g., to staff, to stakeholders, to an external audience)	2.61	3.00	153	1.278
Developing future project plans based on evaluation results	3.07	3.00	151	1.123

Motivation for Involvement – ATE

The questions related to motivation asked respondents to choose an option (required, asked, or volunteered, was not involved) that best described the motivation for involvement during the specific evaluation stage (planning, implementing, or communication of findings).

Table 17. ATE Motivation – Planning

Required	Asked	Volunteered	N
33.1%	34.6%	32.3%	127

Table 18. ATE Motivation – Implementation

Required	Asked	Volunteered	N
33.6%	34.3%	32.1%	134

Table 19. ATE Motivation – Communication

Required	Asked	Volunteered	N
36.8%	28.6%	34.6%	133

Influence on the Evaluation – ATE

Questions 17 – 22 asked about the perception of influence on the program evaluation, differentiating between consideration and actual use.

Table 20. ATE Influence on the Evaluation

Items 17-22	**Was not involved**	**No**	**Yes, a little**	**Yes, some**	**Yes, extensively**	**Do not know**	**N**
Input during *planning* was taken into consideration	18.0%	.6%	6.8%	30.4%	40.4%	3.7%	161
Input during *implementation* was taken into consideration	17.5%	.0%	8.8%	28.8%	38.8%	6.3%	160
Input during *communication* was taken into consideration	16.5%	1.3%	13.3%	26.6%	38.0%	4.4%	158
Input during *planning* was used.	17.1%	1.3%	12.0%	28.5%	35.4%	5.7%	158
Input during *implementation* was used.	16.7%	.6%	11.5%	28.2%	34.6%	8.3%	156
Input during *communication* was used.	15.4%	.6%	14.1%	24.4%	38.5%	7.1%	156

Evaluation Influence – ATE

The survey asked respondents to reflect on the extent of the influence the program evaluation had on their knowledge/understanding (Q25 - Q30), skills (Q31 – Q36), and beliefs (Q38 – Q43). Question 37 asked whether the evaluation had a negative impact and then offered space for a written explanation. The results for this question are presented before the Influence on Beliefs table.

Table 21. ATE Influence on Knowledge & Understanding

Items 25-30	**No**	**Yes, a little**	**Yes, some**	**Yes, extensively**	**N**
The evaluation increased my knowledge/ understanding of how to plan an evaluation.	10.9%	23.7%	44.9%	20.5%	156
The evaluation increased my knowledge/ understanding of how to implement an evaluation.	11.0%	25.8%	43.9%	19.4%	155
The evaluation increased my knowledge/ understanding of how to communicate evaluation findings.	12.3%	25.3%	42.2%	20.1%	154
The evaluation increased my knowledge/ understanding of STEM education.	22.7%	27.9%	35.1%	14.3%	154
The evaluation increased my knowledge/ understanding of STEM education evaluation.	14.2%	27.7%	43.2%	14.8%	155
The evaluation increased my knowledge/ understanding of my project.	11.7%	18.2%	39.0%	31.2%	154

Table 22. ATE Influence on Knowledge/Understanding Means

Items 25-30	Mean	Med	N	SD
The evaluation increased my knowledge/ understanding of how to plan an evaluation.	2.75	3.00	156	.906
The evaluation increased my knowledge/ understanding of how to implement an evaluation.	2.72	3.00	155	.903
The evaluation increased my knowledge/ understanding of how to communicate evaluation findings.	2.70	3.00	154	.930
The evaluation increased my knowledge/ understanding of STEM education.	2.41	2.00	154	.994
The evaluation increased my knowledge /understanding of STEM education evaluation.	2.59	3.00	155	.910
The evaluation increased my knowledge/ understanding of my project.	2.90	3.00	154	.978
Influence on Knowledge/Understanding Mean	2.68	2.83	153	.824

Table 23. ATE Influence on Skills

Items 31-36	No	Yes, a little	Yes, some	Yes, extensively	N
The evaluation improved my skills in planning an evaluation (e.g., discussing the focus of the evaluation, identifying evaluation planning team members, developing the evaluation plan).	16.1%	18.7%	46.5%	18.7%	155
The evaluation improved my skills in implementing an evaluation (e.g., developing data collection instruments and processes, collecting, analyzing, reviewing, and interpreting data).	14.8%	23.9%	43.9%	17.4%	155
The evaluation improved my skills in communicating evaluation findings (e.g., writing and reviewing evaluation reports, presenting evaluation findings).	14.2%	27.7%	40.6%	17.4%	155
The evaluation improved my skills as a STEM educator.	22.6%	28.4%	34.2%	14.8%	155
The evaluation improved my skills as a STEM education evaluator.	19.0%	28.8%	37.3%	15.0%	153
The evaluation improved my skills for working on my project.	14.8%	23.9%	38.7%	22.6%	155

Table 24. ATE Influence on Skills Mean

Items 31-36	Mean	Med	N	SD
The evaluation improved my skills in planning an evaluation.	2.68	3.00	155	.960
The evaluation improved my skills in implementing an evaluation.	2.64	3.00	155	.939
The evaluation improved my skills in communicating evaluation findings.	2.61	3.00	155	.935
The evaluation improved my skills as a STEM educator.	2.41	2.00	155	.999
The evaluation improved my skills as a STEM education evaluator.	2.48	3.00	153	.967
The evaluation improved my skills for working on my project.	2.69	3.00	155	.984
Influence on Skills Mean	2.59	2.75	152	.875

Table 25. ATE Negative Influence on Beliefs

Item	No	Yes	N
Has the evaluation had a negative impact on your belief about evaluation?	96.6%	3.4%	148

Negative Impacts – ATE

Table 26. Examples of negative impact

Item	*Selected Open-Ended Responses* **(10/163 or 6% responded.)**
Examples of Negative Impacts	The evaluation was part of the program so we did it! Although I believe in CPI. The evaluation was marginally useful.
	The data that was sought did not add value.
	The initial results and report of the evaluators was exceedingly academic and not really focused on the project. I discussed this at length with the evaluators and a substantially rewritten, more focused evaluation report was then written.
	I found the evaluation questions off target for the goals of our project.

Influence on Beliefs – ATE

Table 27. Influence on Beliefs

Items 38-43	**No**	**Yes, a little**	**Yes, some**	**Yes, extensively**	**N**
The evaluation increased my belief in the importance of planning an evaluation (e.g., discussing the focus of the evaluation, identifying evaluation planning team members, developing the evaluation plan).	12.2%	17.3%	41.7%	28.8%	156
The evaluation increased my belief in the importance of implementing an evaluation (e.g., developing data collection instruments and processes, collecting, analyzing, reviewing, and interpreting data).	12.8%	21.2%	36.5%	29.5%	156
The evaluation increased my belief in the importance of communicating evaluation findings (e.g., developing future plans for your project, writing and reviewing evaluation reports, presenting evaluation findings).	13.1%	16.3%	45.8%	24.8%	153
The evaluation increased my belief in the importance of STEM education.	27.7%	25.2%	27.1%	20.0%	155
The evaluation increased my belief in the importance of STEM education evaluation.	19.5%	25.3%	31.8%	23.4%	154
The evaluation increased my belief in the importance of my project.	16.9%	20.1%	37.7%	25.3%	154

Table 28. ATE Influence on Beliefs Mean

Items 38-43	**Mean**	**Med**	**N**	**SD**
The evaluation increased my belief in the importance of planning an evaluation.	2.87	3.00	156	.969
The evaluation increased my belief in the importance of implementing an evaluation.	2.83	3.00	156	.998
The evaluation increased my belief in the importance of communicating evaluation findings.	2.82	3.00	153	.954
The evaluation increased my belief in the importance of STEM education.	2.39	2.00	155	1.096
The evaluation increased my belief in the importance of STEM education evaluation.	2.59	3.00	154	1.052
The evaluation increased my belief in the importance of my project.	2.71	3.00	154	1.027
Influence on Beliefs Mean	2.72	3.00	150	.886

Use of Knowledge, Products, and Processes – ATE

The survey directed only respondents who had participated in a subsequent evaluation to respond to questions related to using the evaluation knowledge, products, or processes. If a respondent had not participated in another evaluation, they were instructed to continue with question 50.

Table 29. ATE Use (Knowledge, Products, Processes) in another Evaluation

Items 45-49	**No**	**Yes, a little**	**Yes, some**	**Yes, extensively**	**N**
I used what I learned from planning the program evaluation in another evaluation.	12.2%	24.4%	43.9%	19.5%	41
I used the evaluation plan from the program evaluation as a model in another evaluation.	34.1%	22.0%	29.3%	14.6%	41
I used what I learned from implementing the program evaluation in another evaluation.	17.1%	29.3%	36.6%	17.1%	41
I used data collection instruments from the program evaluation in another evaluation.	36.6%	22.0%	29.3%	12.2%	41
I used what I learned from communicating evaluation findings from the program evaluation in another evaluation.	26.8%	24.4%	31.7%	17.1%	41

Table 30. Use (Knowledge, Products, Processes) Mean in another Evaluation

Items 45-49	**Mean**	**Median**	**N**	**SD**
I used what I learned from planning the program evaluation in another evaluation.	2.71	3.00	41	.929
I used the evaluation plan from the program evaluation as a model in another evaluation.	2.24	2.00	41	1.090
I used what I learned from implementing the program evaluation in another evaluation.	2.54	3.00	41	.977
I used data collection instruments from the program evaluation in another evaluation	2.17	2.00	41	1.070
I used what I learned from communicating evaluation findings from the program evaluation in another evaluation.	2.39	2.00	41	1.070
Use (Knowledge, Products, Processes) Mean in Another Evaluation	2.41	2.20	41	.907

Use of Findings – ATE

The following eight questions (Q50 – Q58) referred to use of the program evaluation findings by project PI's and co-PI's.

Table 31. Use of Findings

Items 50-58	**No**	**Yes, a little**	**Yes, some**	**Yes, extensively**	**N**
I used the program evaluation findings to make decisions about the future existence of my project (e.g., cancel or continue).	25.0%	14.7%	32.1%	28.2%	156
I used the program evaluation findings to develop future plans for my project.	15.4%	12.8%	36.5%	35.3%	156
I used the program evaluation findings to increase the attention given to my project.	17.3%	19.2%	36.5%	26.9%	156
I used the program evaluation findings to meet contractual and/or legal requirements of my project.	25.2%	16.1%	32.9%	25.8%	155
I used the program evaluation findings to enhance my organization's commitment to my project.	28.4%	20.0%	33.5%	18.1%	155
I used the program evaluation findings to seek funding for my project(s).	31.6%	14.8%	31.0%	22.6%	155
I used the program evaluation findings to promote additional evaluation of my project(s).	45.2%	11.6%	27.1%	16.1%	155
I used the program evaluation findings to make changes to my project(s).	17.8%	15.9%	43.3%	22.9%	157
I used the program evaluation findings to help plan a new project(s).	24.5%	13.5%	36.8%	25.2%	155

Table 32. Use of Findings Mean

Items 50-58	**Mean**	**Median**	**N**	**SD**
I used the program evaluation findings to make decisions about the future existence of my project (e.g., cancel, continue).	2.63	3.00	156	1.142
I used the program evaluation findings to develop future plans for my project.	2.92	3.00	156	1.047
I used the program evaluation findings to increase the attention given to my project.	2.73	3.00	156	1.043
I used the program evaluation findings to meet contractual and/or legal requirements of my project.	2.59	3.00	155	1.126
I used the program evaluation findings to enhance my organization's commitment to my project.	2.41	3.00	155	1.086
I used the program evaluation findings to seek funding for my project(s).	2.45	3.00	155	1.157
I used the program evaluation findings to promote additional evaluation of my project(s).	2.14	2.00	155	1.165
I used the program evaluation findings to make changes to my project(s).	2.71	3.00	157	1.013
I used the program evaluation findings to help plan a new project(s).	2.63	3.00	155	1.112
Use of Findings Mean	2.58	2.67	150	.877

Influence of Evaluation on Larger Community – ATE

Table 33. ATE Influence on STEM Community

Not influential at all	Marginally influential	Somewhat influential	Influential	Very influential	Total
5.9%	23.0%	36.8%	26.3%	7.9%	152

Table 34. ATE Influence on Evaluation Community

Not influential at all	Marginally influential	Somewhat influential	Influential	Very influential	Total
8.7%	28.7%	36.7%	17.3%	8.7%	150

Facilitators and Barriers to Influence – ATE

Table 35. Factors Enabling Evaluation Influence

Item	***Selected Open-Ended Responses* (36% or 58/163 responded)**
What major factors enabled the evaluation experience to influence you?	The findings. They confirmed what we suspected and gave us hard evidence we could share with others who may or may not have had the same suspicions we had prior to the evaluation
	The evaluation findings and its communication helped me to enhance my organization and NSF commitment to my project. The evaluation results and reports helped to educate the administrators who their majority do not have an STEM background and education, teaching experience
	Frequent interaction and discussion with my outside evaluator and Co-PIs
	Meeting with the evaluator to discuss findings and related issues helped to focus better on the goals of the project. Reviewing the final report through several versions gave insight to how the project was perceived from the outside and how that perception differed from the internal intent.
	The evaluator was very experienced and skilled and provided an excellent third party overview of our ATE project and provided very appropriate and useful recommendations.
	Our first review reflected a scattered approach to achieving our project's goal. This review stated we needed to locate an experienced External Evaluators and discuss the evaluation role in the project. Also it recommended we hire a Project Coordinator to work under our PI to keep the project organized. Those suggestions pulled our project back on track and it was highly successful.
	Increased my awareness of where other evaluators placed their emphasis and pointed out some evaluation areas I need to further pursue.
	Mainly it was the bringing together of the multiple participants in the project that enabled the evaluation experience to influence me. The project is rather large and multifaceted, and although interrelated, we do not frequently get a chance to see what each other are doing (on a detailed level I mean). The evaluation helped bridge this detailed knowledge gap.

Table 36. Factors Limiting Evaluation Influence

Item	*Selected Open-Ended Responses* **(28% or 45/163responded)**
What major factors limited the influence the evaluation experience had on you?	Too little time to complete all of the evaluation components we envisioned/wished for.
	The process became very drawn out and lost some of its impact over time; the evaluator was often difficult to coordinate with, in terms of matching schedules and common agreement on techniques to use.
	One major limitation was insuring that participating teachers encouraged their students to participate in both pre- and post-tests. A second factor was finding assessment instruments that were sensitive enough to track learning.
	The evaluation survey does not have the capability to capture the products produced by my organization, which includes promotional items, books that highlight ATE, etc. as they are not considered curriculum or materials development.
	Many of the categories were so ambiguous. I felt it was hard to answer correctly. I wish there was a sample survey session to go through at ATE in October to go through so that first time PIs knew how to prepare information for the survey and also so PIs that have done it could be in a session with other PIs to hear their issues with the survey. I believe this would help give some consistency to how the survey is understood
	The questions did not seem relevant to my project.
	Time and money. We should have allocated more funds for evaluation. This did happen in the new project (ITEST).

Inappropriate Use - ATE

Table 37. ATE Inappropriate Use

Item	No	Yes	N
Are you aware of any inappropriate use of the ATE program evaluation?	99.4%	.6%	155

Table 38. ATE Inappropriate Use

Item	*Open-Ended Response*
Comment on identified inappropriate evaluation use.	Project evaluations often focus only on student satisfaction surveys because of the lack of proper training in evaluation and educational research methods.

Table 39. ATE General Comments

Item	*Open-Ended Responses*	N	%
Is there anything else you would like to share about the program evaluation?	No comment.	112	68.7%
	Commented about the evaluation.	29	17.8%
	Commented about the survey.	9	5.5%
	Referred to another open-ended item.	1	0.6%
	Wrote in “none, no, or no comment”.	12	7.4%
Total		163	100.0%

MSP Responses

Table 40. Surveys Completed

Respondents	Response Rate		Removed from Sample
41	48%	41/85	17

Table 41. Evaluation Experience – MSP

First Evaluation	Novice	Skilled	Expert	Total
3	16	20	2	41
7.3%	39.0%	48.8%	4.9%	100.0%

Involvement – MSP

Thirteen of the first 16 questions on the survey asked respondents about their perceptions of involvement during the planning, implementation and communication phases of the program evaluation. Each involvement question began with "I was involved in…." This section included a question at the end of each phase regarding the motivation for involvement. Results of the motivation questions (4, 11, & 16) are presented after the Involvement tables.

Table 42. MSP Involvement – Planning

Items 1-3	**No**	**Yes, A little**	**Yes, Some**	**Yes, Extensively**	**N**
Discussions that focused the evaluation	22.5%	10.0%	17.5%	50.0%	40
Identifying evaluation planning team members	37.8%	10.8%	21.6%	29.7%	37
Developing the evaluation plan	30.8%	12.8%	25.6%	30.8%	39

Table 43. MSP Involvement – Implementation

Items 5-10	**No**	**Yes, A little**	**Yes, Some**	**Yes, Extensively**	**N**
Developing data collection instruments	52.5%	17.5%	17.5%	12.5%	40
Developing data collection processes	45.0%	30.0%	15.0%	10.0%	40
Collecting data	45.0%	17.5%	25.0%	12.5%	40
Reviewing collected data for accuracy and/or completeness	55.0%	20.0%	12.5%	12.5%	40
Analyzing data	43.6%	25.6%	15.4%	15.4%	39
Interpreting collected data	30.0%	22.5%	22.5%	25.0%	40

Table 44. MSP Involvement – Communications

Items 12-15	**No**	**Yes, a little**	**Yes, some**	**Yes, extensively**	**N**
Writing evaluation reports	59.0%	17.9%	15.4%	7.7%	39
Reviewing evaluation reports for accuracy and/or completeness	28.9%	23.7%	18.4%	28.9%	38
Presenting evaluation findings (e.g., to staff, to stakeholders, to an external audience)	38.5%	25.6%	20.5%	15.4%	39
Developing future project plans based on evaluation results	23.1%	20.5%	12.8%	43.6%	39

Table 45. Involvement – MSP

Items 1-3, 5-10, 12-15	Mean	Median	N	SD
Discussions that focused the evaluation	2.95	3.50	40	1.239
Identifying evaluation planning team members	2.43	3.00	37	1.281
Developing the evaluation plan	2.56	3.00	39	1.231
Developing data collection instruments	1.90	1.00	40	1.105
Developing data collection processes	1.90	2.00	40	1.008
Collecting data	2.05	2.00	40	1.108
Reviewing collected data for accuracy and/or completeness	1.83	1.00	40	1.083
Analyzing data	2.03	2.00	39	1.112
Interpreting collected data	2.43	2.00	40	1.174
Writing evaluation reports	1.72	1.00	39	.999
Reviewing evaluation reports for accuracy and/or completeness	2.47	2.00	38	1.202
Presenting evaluation findings (e.g., to staff, to stakeholders, to an external audience)	2.13	2.00	39	1.105
Developing future project plans based on evaluation results	2.77	3.00	39	1.245

Motivation for Involvement – MSP

The questions related to motivation asked respondents to choose an option (required, asked, or volunteered, was not involved) that best described the motivation for involvement during the specific evaluation stage (planning, implementing, or communication of findings).

Table 46. MSP Motivation – Planning

Required	Asked	Volunteered	N
32.1%	28.6%	39.3%	28

Table 47. MSP Motivation – Implementation

Required	Asked	Volunteered	N
32.3%	19.4%	48.4%	31

Table 48. MSP Motivation – Communication

Required	Asked	Volunteered	N
39.3%	3.6%	57.1%	28

Influence on the Evaluation – MSP

Questions 17 – 22 asked about the perception of influence on the program evaluation, differentiating between consideration and actual use.

Table 49. MSP Influence on the Evaluation

Items 17-22	**Was not involved**	**No**	**Yes, a little**	**Yes, some**	**Yes, extensively**	**Do not know**	**N**
Input during *planning* was taken into consideration	25.0%	.0%	12.5%	27.5%	32.5%	2.5%	40
Input during *implementation* was taken into consideration	26.8%	.0%	14.6%	24.4%	29.3%	4.9%	41
Input during *communication* was taken into consideration	24.4%	.0%	19.5%	24.4%	26.8%	4.9%	41
Input during *planning* was used.	24.4%	.0%	12.2%	24.4%	34.1%	4.9%	41
Input during *implementation* was used.	24.4%	2.4%	14.6$	26.8%	26.8%	4.9%	41
Input during *communication* was used.	24.4%	.0%	22.0%	22.0%	24.4%	7.3%	41

Evaluation Influence – MSP

The survey asked respondents to reflect on the extent of the influence the program evaluation had on their knowledge/understanding (Q25 - Q30), skills (Q31 – Q36), and beliefs (Q38 – Q43). Question 37 asked whether the evaluation had a negative impact and then offered space for a written explanation. The results for this question are presented before the Influence on Beliefs table.

Table 50. MSP Influence on Knowledge & Understanding

Items 25-30	**No**	**Yes, a little**	**Yes, some**	**Yes, extensively**	**N**
The evaluation increased my knowledge/ understanding of how to plan an evaluation.	22.0%	12.2%	39.0%	26.8%	41
The evaluation increased my knowledge/ understanding of how to implement an evaluation.	19.5%	12.2%	46.3%	22.0%	41
The evaluation increased my knowledge/ understanding of how to communicate evaluation findings.	22.0%	19.5%	39.0%	19.5%	41
The evaluation increased my knowledge/ understanding of STEM education.	30.0%	22.5%	35.0%	12.5%	40
The evaluation increased my knowledge/ understanding of STEM education evaluation.	19.5%	14.6%	43.9%	22.0%	41
The evaluation increased my knowledge/ understanding of my project.	19.5%	9.8%	34.1%	36.6%	41

Table 51. MSP Influence on Knowledge/Understanding Means

Items 25-30	Mean	Med	N	SD
The evaluation increased my knowledge/ understanding of how to plan an evaluation.	2.71	3.00	41	1.101
The evaluation increased my knowledge/ understanding of how to implement an evaluation.	2.71	3.00	41	1.031
The evaluation increased my knowledge/ understanding of how to communicate evaluation findings.	2.56	3.00	41	1.050
The evaluation increased my knowledge/ understanding of Science, Technology, Engineering, and Mathematics (STEM) education.	2.30	2.00	40	1.043
The evaluation increased my knowledge/ understanding of STEM education evaluation.	2.68	3.00	41	1.035
The evaluation increased my knowledge/ understanding of my project.	2.88	3.00	41	1.122
Influence on Knowledge/Understanding Mean	2.61	2.83	40	.933

Table 52. MSP Influence on Skills

Items 31-36	No	Yes, a little	Yes, some	Yes, extensively	N
The evaluation improved my skills in planning an evaluation (e.g., discussing the focus of the evaluation, identifying evaluation planning team members, developing the evaluation plan).	17.5%	17.5%	47.5%	17.5%	40
The evaluation improved my skills in implementing an evaluation (e.g., developing data collection instruments and processes, collecting, analyzing, reviewing, and interpreting data).	19.5%	29.3%	39.0%	12.2%	41
The evaluation improved my skills in communicating evaluation findings (e.g., writing and reviewing evaluation reports, presenting evaluation findings).	22.0%	31.7%	39.0%	7.3%	41
The evaluation improved my skills as a STEM educator.	36.6%	24.4%	26.8%	12.2%	41
The evaluation improved my skills as a STEM education evaluator.	17.1%	29.3%	43.9%	9.8%	41
The evaluation improved my skills for working on my project.	22.0%	14.6%	43.9%	19.5%	41

Table 53. MSP Influence on Skills Mean

Items 31-36	Mean	Med	N	SD
The evaluation improved my skills in planning an evaluation.	2.65	3.00	40	.975
The evaluation improved my skills in implementing an evaluation.	2.44	3.00	41	.950
The evaluation improved my skills in communicating evaluation findings.	2.32	2.00	41	.907
The evaluation improved my skills as a STEM educator.	2.15	2.00	41	1.062
The evaluation improved my skills as a STEM education evaluator.	2.46	3.00	41	.897
The evaluation improved my skills for working on my project.	2.61	3.00	41	1.046
Influence on Skills Mean	2.47	2.67	40	.834

Table 54. MSP Negative Influence on Beliefs

Item	No	Yes	N
Has the evaluation had a negative impact on your belief about evaluation?	95.1%	4.9%	41

Negative Impacts – MSP

Table 55. Examples of negative impact

Item	*Selected Open-Ended Responses* (4/41 or 10% responded.)
Examples of Negative Impacts	Doing the work doesn't mean the project director will value it
	We provided our evaluators with a detailed set of questions we wanted addressed, and it appeared that they thought the questions were reasonable. In practice, however, their insistence on only using normed and validated instruments was greatly limiting, driving the focus off from the questions we were interested and more to the questions they felt comfortable addressing. Also, as a research project proceeds there are many adjustments made to optimize progress. This causes an inherent tension between the optimal execution of the project and the optimal (i.e., stable and predictable) conditions for evaluation.

Influence on Beliefs – MSP

Table 56. Influence on Beliefs

Items 38-43	No	Yes, a little	Yes, some	Yes, extensively	N
The evaluation increased my belief in the importance of planning an evaluation (e.g., discussing the focus of the evaluation, identifying evaluation planning team members, developing the evaluation plan).	22.0%	12.2%	29.3%	36.6%	41
The evaluation increased my belief in the importance of implementing an evaluation (e.g., developing data collection instruments and processes, collecting, analyzing, reviewing, and interpreting data).	22.0%	12.2%	29.3%	36.6%	41
The evaluation increased my belief in the importance of communicating evaluation findings (e.g., developing future plans for your project, writing and reviewing evaluation reports, presenting evaluation findings).	31.7%	9.8%	22.0%	36.6%	41
The evaluation increased my belief in the importance of STEM education.	48.8%	17.1%	22.0%	12.2%	41
The evaluation increased my belief in the importance of STEM education evaluation.	29.3%	9.8%	43.9%	17.1%	41
The evaluation increased my belief in the importance of my project.	24.4%	17.1%	34.1%	24.4%	41

Table 57. MSP Influence on Beliefs Mean

Items 38-43	**Mean**	**Med**	**N**	**SD**
The evaluation increased my belief in the importance of planning an evaluation.	2.80	3.00	41	1.167
The evaluation increased my belief in the importance of implementing an evaluation.	2.80	3.00	41	1.167
The evaluation increased my belief in the importance of communicating evaluation findings.	2.63	3.00	41	1.280
The evaluation increased my belief in the importance of STEM education.	1.98	2.00	41	1.107
The evaluation increased my belief in the importance of STEM education evaluation.	2.49	3.00	41	1.098
The evaluation increased my belief in the importance of my project.	2.59	3.00	41	1.117
Influence on Beliefs Mean	2.55	2.83	41	1.014

Use of Knowledge, Products, and Processes – MSP

The survey directed only respondents who had participated in a subsequent evaluation to respond to questions related to using the evaluation knowledge, products, or processes. If a respondent had not participated in another evaluation, they were instructed to continue with question 50.

Table 58. MSP Use (Knowledge, Products, Processes) in another Evaluation

Items 45-49	**No**	**Yes, a little**	**Yes, some**	**Yes, extensively**	**N**
I used what I learned from planning the program evaluation in another evaluation.	18.2%	13.6%	27.3%	40.9%	22
I used the evaluation plan from the program evaluation as a model in another evaluation.	22.7%	18.2%	27.3%	31.8%	22
I used what I learned from implementing the program evaluation in another evaluation.	18.2%	22.7%	18.2%	40.9%	22
I used data collection instruments from the program evaluation in another evaluation.	31.8%	13.6%	36.4%	18.2%	22
I used what I learned from communicating evaluation findings from the program evaluation in another evaluation.	27.3%	18.2%	31.8%	22.7%	22

Table 59. Use (Knowledge, Products, Processes) Mean in another Evaluation

Items 45-49	**Mean**	**Median**	**N**	**SD**
I used what I learned from planning the program evaluation in another evaluation.	2.91	3.00	22	1.151
I used the evaluation plan from the program evaluation as a model in another evaluation.	2.68	3.00	22	1.171
I used what I learned from implementing the program evaluation in another evaluation.	2.82	3.00	22	1.181
I used data collection instruments from the program evaluation in another evaluation	2.41	3.00	22	1.141
I used what I learned from communicating evaluation findings from the program evaluation in another evaluation.	2.50	3.00	22	1.144
Use (Knowledge, Products, Processes) Mean in Another Evaluation	2.66	2.90	22	1.070

Use of Findings – MSP

The following eight questions (Q50 – Q58) referred to use of the program evaluation findings by project PI's and co-PI's.

Table 60. Use of Findings

Items 50-58	**No**	**Yes, a little**	**Yes, some**	**Yes, extensively**	**N**
I used the program evaluation findings to make decisions about the future existence of my project (e.g., cancel or continue).	29.3%	9.8%	36.6%	24.4%	41
I used the program evaluation findings to develop future plans for my project.	24.4%	9.8%	36.6%	29.3%	41
I used the program evaluation findings to increase the attention given to my project.	34.1%	12.2%	31.7%	22.0%	41
I used the program evaluation findings to meet contractual and/or legal requirements of my project.	34.1%	7.3%	24.4%	34.1%	41
I used the program evaluation findings to enhance my organization's commitment to my project.	34.1%	14.6%	34.1%	17.1%	41
I used the program evaluation findings to seek funding for my project(s).	36.6%	4.9%	36.6%	22.0%	41
I used the program evaluation findings to promote additional evaluation of my project(s).	39.0%	9.8%	29.3%	22.0%	41
I used the program evaluation findings to make changes to my project(s).	29.3%	4.9%	36.6%	29.3%	41
I used the program evaluation findings to help plan a new project(s).	31.7%	9.8%	36.6%	22.0%	41

Table 61. Use of Findings Mean

Items 50-58	**Mean**	**Median**	**N**	**SD**
I used the program evaluation findings to make decisions about the future existence of my project (e.g., cancel, continue).	2.56	3.00	41	1.163
I used the program evaluation findings to develop future plans for my project.	2.71	3.00	41	1.146
I used the program evaluation findings to increase the attention given to my project.	2.41	3.00	41	1.183
I used the program evaluation findings to meet contractual and/or legal requirements of my project.	2.59	3.00	41	1.284
I used the program evaluation findings to enhance my organization's commitment to my project.	2.34	3.00	41	1.132
I used the program evaluation findings to seek funding for my project(s).	2.44	3.00	41	1.205
I used the program evaluation findings to promote additional evaluation of my project(s).	2.34	3.00	41	1.217
I used the program evaluation findings to make changes to my project(s).	2.66	3.00	41	1.196
I used the program evaluation findings to help plan a new project(s).	2.49	3.00	41	1.165
Use of Findings Mean	2.50	2.78	41	1.027

Influence of Evaluation on Larger Community – MSP

Table 62. MSP Influence on STEM Community

Not influential at all	Marginally influential	Somewhat influential	Influential	Very influential	Total
12.5%	22.5%	37.5%	22.5%	5.0%	40

Table 63. MSP Influence on Evaluation Community

Not influential at all	Marginally influential	Somewhat influential	Influential	Very influential	Total
17.5%	30.0%	30.0%	17.5%	5.0%	40

Facilitators and Barriers to Influence – MSP

Table 64. Factors Enabling Evaluation Influence

Item	***Selected Open-Ended Responses* (63% or 26/41 responded)**
What major factors enabled the evaluation experience to influence you?	Very carefully designed and reviewed as we went along - very introspective.
	Expert evaluators led me through the process from logic models through data collection, to analysis and crafting reports and communications.
	The open communication between the evaluation team and PIs. / The mutual respect between the evaluators and PIs
	Discussions and back and forth communications with the evaluators were quite helpful. I had not expected the formative part of the evaluation to be as helpful as it was.
	It was the first time I had learned about evaluation and I found it interesting. Perhaps most importantly, I was in seminars about Formative Assessment that I then put to use in my own teaching. The positive experience I had with Formative Assessment oriented me towards assessments of all types. I attended presentations on evaluation data given by very experienced people in our project and was able to learn from them.
	Emphasis of NSF; support of program officer
	The experience allowed me to use evaluation results to plan and revise program activities, and to "market" the MSP to other at the local and state level.

Table 65. Factors Limiting Evaluation Influence

Item	*Selected Open-Ended Responses* **(37% or 15/41 responded)**
What major factors limited the influence the evaluation experience had on you?	I was the PI and Project Director--so I was very busy with many aspects of the project and did not have full time to spend on evaluation or learning how to do it. That is why I hired evaluators--to do the work. I wished I could do more with them, but there was no time.
	A frustration was trying to balance the need for established and validated instruments with the need for instruments that were a good match for our work. We opted for established instruments which did work reasonably well, but I felt this was a limitation.
	Mobility of teachers and students; difficulty in managing data
	I collected and supplied data for my project as requested for the overall program evaluation. We also collected, analyzed, and reported data specific to our individual project. The latter, local evaluation has been very important to the project. The MSP Program-wide evaluation hasn't had much, if any influence on our work. I'm not sure I'm aware of the findings. If I were, they might be influential.
	Lack of understanding in the project team of the uses of evaluation for stakeholders apart from a research perspective, and the need for timely feedback to teachers in particular.

Inappropriate Use - MSP

Table 66. MSP Inappropriate Use

Item	No	Yes	N
Are you aware of any inappropriate use of the MSP program evaluation?	100%	.0%	40

Table 67. MSP General Comments

Item	*Open-Ended Responses*	N	%
Is there anything else you would like to share about the program evaluation?	No comment.	26	63.4%
	Commented about the evaluation.	12	29.3%
	Commented about the survey.	2	4.9%
	Referred to another open-ended item.	0	0.0%
	Wrote in “none, no, or no comment”.	1	2.4%
Total		41	100.0%

Noyce Responses

Table 68. Surveys Completed

Respondents	Response Rate		Removed from Sample
96	52%	96/185	25

Table 69. Evaluation Experience – Noyce

First Evaluation	Novice	Skilled	Expert	Total
17	31	42	5	95
17.9%	32.6%	44.2%	5.3%	100.0%

Involvement – Noyce

Thirteen of the first 16 questions on the survey asked respondents about their perceptions of involvement during the planning, implementation and communication phases of the program evaluation. Each involvement question began with "I was involved in…." This section included a question at the end of each phase regarding the motivation for involvement. Results of the motivation questions (4, 11, & 16) are presented after the Involvement tables.

Table 70. Noyce Involvement – Planning

Items 1-3	**No**	**Yes, A little**	**Yes, Some**	**Yes, Extensively**	**N**
Discussions that focused the evaluation	27.7%	23.4%	23.4%	25.5%	94
Identifying evaluation planning team members	48.9%	9.6%	16.0%	25.5%	94
Developing the evaluation plan	41.9%	18.3%	17.2%	22.6%	93

Table 71. Noyce Involvement – Implementation

Items 5-10	**No**	**Yes, A little**	**Yes, Some**	**Yes, Extensively**	**N**
Developing data collection instruments	59.1%	15.1%	12.9%	12.9%	93
Developing data collection processes	54.8%	18.3%	10.8%	16.1%	93
Collecting data	47.3%	20.9%	7.7%	24.2%	91
Reviewing collected data for accuracy and/or completeness	49.5%	16.5%	16.5%	17.6%	91
Analyzing data	66.3%	9.8%	9.8%	14.1%	92
Interpreting collected data	52.2%	15.2%	16.3%	16.3%	92

Table 72. Noyce Involvement – Communications

Items 12-15	**No**	**Yes, a little**	**Yes, some**	**Yes, extensively**	**N**
Writing evaluation reports	70.0%	6.7%	11.1%	12.2%	90
Reviewing evaluation reports for accuracy and/or completeness	50.0%	18.9%	13.3%	17.8%	90
Presenting evaluation findings (e.g., to staff, to stakeholders, to an external audience)	67.4%	5.6%	5.6%	21.3%	89
Developing future project plans based on evaluation results	52.7%	11.0%	9.9%	26.4%	91

Table 73. Involvement - Noyce

Items 1-3, 5-10, 12-15	Mean	Median	N	SD
Discussions that focused the evaluation	2.47	2.00	94	1.152
Identifying evaluation planning team members	2.18	2.00	94	1.287
Developing the evaluation plan	2.20	2.00	93	1.212
Developing data collection instruments	1.80	1.00	93	1.099
Developing data collection processes	1.88	1.00	93	1.141
Collecting data	2.09	2.00	91	1.235
Reviewing collected data for accuracy and/or completeness	2.02	2.00	91	1.174
Analyzing data	1.72	1.00	92	1.122
Interpreting collected data	1.97	1.00	92	1.162
Writing evaluation reports	1.66	1.00	90	1.093
Reviewing evaluation reports for accuracy and/or completeness	1.99	1.50	90	1.166
Presenting evaluation findings (e.g., to staff, to stakeholders, to an external audience)	1.81	1.00	89	1.251
Developing future project plans based on evaluation results	2.10	1.00	91	1.300

Motivation for Involvement – Noyce

The questions related to motivation asked respondents to choose an option (required, asked, or volunteered, was not involved) that best described the motivation for involvement during the specific evaluation stage (planning, implementing, or communication of findings).

Table 74. Noyce Motivation – Planning

Required	Asked	Volunteered	N
25.0%	43.8%	31.3%	64

Table 75. Noyce Motivation – Implementation

Required	Asked	Volunteered	N
36.2%	39.1%	24.6%	69

Table 76. Noyce Motivation – Communication

Required	Asked	Volunteered	N
38.8%	26.5%	34.7%	49

Influence on the Evaluation – Noyce

Questions 17 – 22 asked about the perception of influence on the program evaluation, differentiating between consideration and actual use.

Table 77. Noyce Influence on the Evaluation

Items 17-22	**Was not involved**	**No**	**Yes, a little**	**Yes, some**	**Yes, extensively**	**Do not know**	**N**
Input during *planning* was taken into consideration	29.5%	1.1%	17.9%	15.8%	25.3%	10.5%	95
Input during *implementation* was taken into consideration	33.7%	.0%	11.6%	17.9%	24.2%	12.6%	95
Input during *communication* was taken into consideration	44.7%	.0%	7.4%	14.9%	20.2%	12.8%	94
Input during *planning* was used.	30.5%	3.2%	15.8%	17.9%	20.0%	12.6%	95
Input during *implementation* was used.	35.8%	2.1%	15.8%	15.8%	20.0%	10.5%	95
Input during *communication* was used.	43.2%	2.1%	9.5%	13.7%	20.0%	11.6%	95

Evaluation Influence – Noyce

The survey asked respondents to reflect on the extent of the influence the program evaluation had on their knowledge/understanding (Q25 - Q30), skills (Q31 – Q36), and beliefs (Q38 – Q43). Question 37 asked whether the evaluation had a negative impact and then offered space for a written explanation. The results for this question are presented before the Influence on Beliefs table.

Table 78. Noyce Influence on Knowledge & Understanding

Items 25-30	**No**	**Yes, a little**	**Yes, some**	**Yes, extensively**	**N**
The evaluation increased my knowledge/ understanding of how to plan an evaluation.	35.8%	32.6%	27.4%	4.2%	95
The evaluation increased my knowledge/ understanding of how to implement an evaluation.	35.8%	33.7%	27.4%	3.2%	95
The evaluation increased my knowledge/ understanding of how to communicate evaluation findings.	43.2%	24.2%	30.5%	2.1%	95
The evaluation increased my knowledge /understanding of STEM education.	48.4%	28.4%	18.9%	4.2%	95
The evaluation increased my knowledge /understanding of STEM education evaluation.	36.8%	35.8%	23.2%	4.2%	95
The evaluation increased my knowledge/ understanding of my project.	26.3%	27.4%	32.6%	13.7%	95

Table 79. Noyce Influence on Knowledge/Understanding Means

Items 25-30	Mean	Med	N	SD
The evaluation increased my knowledge/ understanding of how to plan an evaluation.	2.00	2.00	95	.899
The evaluation increased my knowledge/ understanding of how to implement an evaluation.	1.98	2.00	95	.875
The evaluation increased my knowledge/ understanding of how to communicate evaluation findings.	1.92	2.00	95	.907
The evaluation increased my knowledge/ understanding of Science, Technology, Engineering, and Mathematics (STEM) education.	1.79	2.00	95	.898
The evaluation increased my knowledge/ understanding of STEM education evaluation.	1.95	2.00	95	.880
The evaluation increased my knowledge/ understanding of my project.	2.34	2.00	95	1.017
Influence on Knowledge/Understanding Mean	1.99	2.00	95	.784

Table 80. Noyce Influence on Skills

Items 31-36	No	Yes, a little	Yes, some	Yes, extensively	N
The evaluation improved my skills in planning an evaluation (e.g., discussing the focus of the evaluation, identifying evaluation planning team members, developing the evaluation plan).	42.1%	28.4%	21.1%	8.4%	95
The evaluation improved my skills in implementing an evaluation (e.g., developing data collection instruments and processes, collecting, analyzing, reviewing, and interpreting data).	47.4%	24.2%	21.1%	7.4%	95
The evaluation improved my skills in communicating evaluation findings (e.g., writing and reviewing evaluation reports, presenting evaluation findings).	51.6%	23.2%	20.0%	5.3%	95
The evaluation improved my skills as a STEM educator.	52.6%	26.3%	18.9%	2.1%	95
The evaluation improved my skills as a STEM education evaluator.	44.2%	31.6%	20.0%	4.2%	95
The evaluation improved my skills for working on my project.	33.0%	38.3%	21.3%	7.4%	94

Table 81. Noyce Influence on Skills Mean

Items 31-36	Mean	Med	N	SD
The evaluation improved my skills in planning an evaluation.	1.96	2.00	95	.988
The evaluation improved my skills in implementing an evaluation.	1.88	2.00	95	.988
The evaluation improved my skills in communicating evaluation findings.	1.79	1.00	95	.944
The evaluation improved my skills as a STEM educator.	1.71	1.00	95	.849
The evaluation improved my skills as a STEM education evaluator.	1.84	2.00	95	.891
The evaluation improved my skills for working on my project.	2.03	2.00	94	.921
Influence on Skills Mean	1.87	1.67	94	.817

Table 82. Noyce Negative Influence on Beliefs

Item	No	Yes	N
Has the evaluation had a negative impact on your belief about evaluation?	95.8%	4.2%	95

Negative Impacts – Noyce

Table 83. Examples of negative impact

Item	*Selected Open-Ended Responses* (7/96 or 7% responded.)
Examples of Negative Impacts	As Principal Investigator, I had relatively little input in the development of Noyce Scholars. It was very frustrating. I had previously had a grant in which I recruited and worked closely with science teacher interns - much more satisfying for me and for them. Noyce provides little support for development; the primary benefit of the grant is to provide financial assistance to students. So the question is, what exactly are we evaluating -- the effectiveness of the credential program?
	There ought to be a more direct methodology. Certainly, the method may be dependent on the objectives of a program.
	I've become concerned about the cost-benefit ratio of an evaluation system that relies on "professionals" who have such a strong vested interest in the process rather than on simple collection of key outcomes.
	All of the Co-Pi's who were supposed to design and implement the evaluation either left the project or refused to do any work. Everything was left to me, yet that was never to be my role in the project. This was a disaster for me and makes me feel guilty about the project. I had no way to make anyone do anything. Eventually, the Dean of the College of Education simply admitted that they had no ability to do what they had promised.

Influence on Beliefs – Noyce

Table 84. Influence on Beliefs

Items 38-43	**No**	**Yes, a little**	**Yes, some**	**Yes, extensively**	**N**
The evaluation increased my belief in the importance of planning an evaluation (e.g., discussing the focus of the evaluation, identifying evaluation planning team members, developing the evaluation plan).	37.2%	26.6%	17.0%	19.1%	94
The evaluation increased my belief in the importance of implementing an evaluation (e.g., developing data collection instruments and processes, collecting, analyzing, reviewing, and interpreting data).	36.6%	30.1%	19.4%	14.0%	93
The evaluation increased my belief in the importance of communicating evaluation findings (e.g., developing future plans for your project, writing and reviewing evaluation reports, presenting evaluation findings).	38.3%	29.8%	18.1%	13.8%	94
The evaluation increased my belief in the importance of STEM education.	48.9%	21.3%	18.1%	11.7%	94
The evaluation increased my belief in the importance of STEM education evaluation.	44.7%	23.4%	20.2%	11.7%	94
The evaluation increased my belief in the importance of my project.	37.2%	25.5%	20.2%	17.0%	94

Table 85. Noyce Influence on Beliefs Mean

Items 38-43	**Mean**	**Med**	**N**	**SD**
The evaluation increased my belief in the importance of planning an evaluation.	2.18	2.00	94	1.136
The evaluation increased my belief in the importance of implementing an evaluation.	2.11	2.00	93	1.058
The evaluation increased my belief in the importance of communicating evaluation findings.	2.07	2.00	94	1.060
The evaluation increased my belief in the importance of STEM education.	1.93	2.00	94	1.070
The evaluation increased my belief in the importance of STEM education evaluation.	1.99	2.00	94	1.063
The evaluation increased my belief in the importance of my project.	2.17	2.00	94	1.113
Influence on Beliefs Mean	2.08	2.00	93	.951

Use of Knowledge, Products, and Processes – Noyce

The survey directed only respondents who had participated in a subsequent evaluation to respond to questions related to using the evaluation knowledge, products, or processes. If a respondent had not participated in another evaluation, they were instructed to continue with question 50.

Table 86. Noyce Use (Knowledge, Products, Processes) in another Evaluation

Items 45-49	**No**	**Yes, a little**	**Yes, some**	**Yes, extensively**	**N**
I used what I learned from planning the program evaluation in another evaluation.	41.0%	28.2%	25.6%	5.1%	39
I used the evaluation plan from the program evaluation as a model in another evaluation.	61.5%	23.1%	12.8%	2.6%	39
I used what I learned from implementing the program evaluation in another evaluation.	41.0%	33.3%	23.1%	2.6%	39
I used data collection instruments from the program evaluation in another evaluation.	79.5%	5.1%	10.3%	5.1%	39
I used what I learned from communicating evaluation findings from the program evaluation in another evaluation.	61.5%	12.8%	23.1%	2.6%	39

Table 87. Use (Knowledge, Products, Processes) Mean in another Evaluation

Items 45-49	**Mean**	**Median**	**N**	**SD**
I used what I learned from planning the program evaluation in another evaluation.	1.95	2.00	39	.944
I used the evaluation plan from the program evaluation as a model in another evaluation.	1.56	1.00	39	.821
I used what I learned from implementing the program evaluation in another evaluation.	1.87	2.00	39	.864
I used data collection instruments from the program evaluation in another evaluation	1.41	1.00	39	.880
I used what I learned from communicating evaluation findings from the program evaluation in another evaluation.	1.67	1.00	39	.927
Use (Knowledge, Products, Processes) Mean in Another Evaluation	1.69	1.60	39	.674

Use of Findings – Noyce

The following eight questions (Q50 – Q58) referred to use of the program evaluation findings by project PI's and co-PI's.

Table 88. Use of Findings

Items 50-58	**No**	**Yes, a little**	**Yes, some**	**Yes, extensively**	**N**
I used the program evaluation findings to make decisions about the future existence of my project (e.g., cancel or continue).	36.7%	26.7%	17.8%	18.9%	90
I used the program evaluation findings to develop future plans for my project.	25.0%	29.3%	25.0%	20.7%	92
I used the program evaluation findings to increase the attention given to my project.	39.6%	20.9%	25.3%	14.3%	91
I used the program evaluation findings to meet contractual and/or legal requirements of my project.	41.3%	14.1%	18.5%	26.1%	92
I used the program evaluation findings to enhance my organization's commitment to my project.	47.8%	14.1%	17.4%	20.7%	92
I used the program evaluation findings to seek funding for my project(s).	52.2%	17.4%	14.1%	16.3%	92
I used the program evaluation findings to promote additional evaluation of my project(s).	58.2%	19.8%	12.1%	9.9%	91
I used the program evaluation findings to make changes to my project(s).	35.9%	32.6%	18.5%	13.0%	92
I used the program evaluation findings to help plan a new project(s).	51.1%	19.6%	17.4%	12.0%	92

Table 89. Use of Findings Mean

Items 50-58	**Mean**	**Median**	**N**	**SD**
I used the program evaluation findings to make decisions about the future existence of my project (e.g., cancel, continue).	2.19	2.00	90	1.131
I used the program evaluation findings to develop future plans for my project.	2.41	2.00	92	1.081
I used the program evaluation findings to increase the attention given to my project.	2.14	2.00	91	1.101
I used the program evaluation findings to meet contractual and/or legal requirements of my project.	2.29	2.00	92	1.254
I used the program evaluation findings to enhance my organization's commitment to my project.	2.11	2.00	92	1.218
I used the program evaluation findings to seek funding for my project(s).	1.95	1.00	92	1.152
I used the program evaluation findings to promote additional evaluation of my project(s).	1.74	1.00	91	1.020
I used the program evaluation findings to make changes to my project(s).	2.09	2.00	92	1.034
I used the program evaluation findings to help plan a new project(s).	1.90	1.00	92	1.080
Use of Findings Mean	2.09	1.89	91	.919

Influence of Evaluation on Larger Community – Noyce

Table 90. Noyce Influence on STEM Community

Not influential at all	Marginally influential	Somewhat influential	Influential	Very influential	Total
14.3%	29.8%	34.5%	13.1%	8.3%	84

Table 91. Noyce Influence on Evaluation Community

Not influential at all	Marginally influential	Somewhat influential	Influential	Very influential	Total
17.7%	46.8%	26.6%	6.3%	2.5%	79

Facilitators and Barriers to Influence – Noyce

Table 92. Factors Enabling Evaluation Influence

Item	*Selected Open-Ended Responses* (29% or 28/96 responded)
What major factors enabled the evaluation experience to influence you?	I enjoy the evaluation process and would have liked to have been more involved in the implementation and dissemination; however, I was not the PI during the course of the project although I was part of the management team. Our project has lost 2 PIs to retirement and now we have a third STEM person. I have enjoyed the Noyce project for the contributions it has made to our STEM students and our university. The evaluation process used in this project has helped me in designing and developing other evaluation strategies for other projects. I like working with NSF and Noyce because the process is straightforward, logical, and easily communicated.
	Good data which was triangulated enabled us to get a good picture of what we do well and what we need to work on. The fact that data collection was done by people outside the project allowed Scholars to freely share what worked for them and what didn't.
	As the PI, I met with the external evaluator on a regular basis, and was provided with insights and suggestions as the project developed.
	Our team work was a driving force and its effect was a motivator for me.
	The evaluator's direct communication and need for my assistance in collecting data about the scholars. (I scheduled times (and locations) for the evaluator to meet with the (busy) scholars (in my absence).) Reading the reports, which are submitted in a timely manner.
	The importance of being accountable to the NSF. The need for feedback in order to make possible adjustments in our procedures of selection and follow-up.
	For the most part, the factor that enabled some influence of the University of Minnesota R3 evaluation data on me was the summary presentation of results at the annual Noyce conferences and the existence of the website where evaluation information is made available.

Table 93. Factors Limiting Evaluation Influence

Item	*Selected Open-Ended Responses* (30% or 29/96 responded)
What major factors limited the influence the evaluation experience had on you?	Time. Work load issues at the university result in limited time to participate in the evaluation experiences of the existing Noyce program
	Difficulty of obtaining teacher participants in evaluation study because of testing pressures.
	It was conducted very professionally. Yet, it did not seem to measure or describe the real strength of the Noyce program: the scholars' potential to influence the lives of STEM students in high need schools. It was too much about numbers.
	Lack of knowledge about and some conflicts about what should be evaluated. STEM is different in its evaluation needs. Those differences need to be honored. Also, I think it is very difficult to measure some of the extremely important issues. Attitudes for example, are difficult to measure.
	The small number of Noyce scholarship recipients. So far we've only awarded 8 scholarships, so I'm not sure if there were any significant trends to evaluate.
	The refusal of co-Pi's to do promised work made it impossible to accomplish this task. I will never work on a project like this again. And, the small sample size, variety of schools and movement of teachers among schools makes the evaluation plan almost worthless.
	I am not involved in the day to day workings of this project; if I had more involvement, I would have learned more.

Inappropriate Use - Noyce

Table 94. Noyce Inappropriate Use

Item	No	Yes	N
Are you aware of any inappropriate use of the Noyce program evaluation?	100%	.0%	91

Table 95. Noyce General Comments

Item	*Open-Ended Responses*	N	%
Is there anything else you would like to share about the program evaluation?	No comment.	68	70.8%
	Commented about the evaluation.	17	17.7%
	Commented about the survey.	6	6.3%
	Referred to another open-ended item.	0	0.0%
	Wrote in “none, no, or no comment”.	5	5.2%
Total		96	100.0%